CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 92-015

WASTE DISCHARGE REQUIREMENTS
FOR

TUOLUMNE MEADOWS WASTEWATER TREATMENT FACILITY
NATIONAL PARK SERVICE
YOSEMITE NATIONAL PARK
TUOLUMNE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Tuolumne Meadows Wastewater Treatment Facility and the National Park Service, (hereafter Discharger) submitted a Report of Waste Discharge and a site evaluation report, both dated 28 February 1991. The property is owned by the National Park Service.

2. Waste Discharge Requirements Order No. 79-44, adopted by the Board on 23 February 1979, prescribes requirements for a discharge from Tuolumne Meadows to an extended aeration treatment plant, two lined ponds, chlorination, and a spray disposal field.

3. Domestic wastewater is generated from an 800-campsite public campground, an 80-bed lodge with a dining room, a general store, a coffee shop, a service station, a ranger station, a visitor information center, employee housing, and RV dumpings.

4. Yosemite National Park annually terminates operations in the area during the winter season. The water supply system is deactivated and the wastewater treatment plant is dismantled to avoid snow load damage. The collection system is flushed to the oxidation ponds prior to allowing infiltration water to be discharged to Tuolumne Meadows during the winter.

5. The liner in the deeper pond was replaced in September 1991 because the old liner had a tear and bubbled up from springs beneath the pond. The new liner has an underdrain system which will drain spring water. This drainage will be monitored for evidence of liner leakage.

6. Order No. 79-44 is neither adequate nor consistent with current plans and policies of the Board.

7. The Discharger discharges an average of 0.12 million gallons per day (mgd) from an extended aeration treatment plant to two lined ponds and a spray disposal field.

8. The facility is in Section 6, T1S, R24E, MDB&M, with surface water drainage to the Tuolumne River, as shown on Attachment A, which is attached hereto and part of the Order by reference.

9. The Board adopted a Water Quality Control Plan, Second Edition, for the San Joaquin River Basin (5C) (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.

10. The beneficial uses of the Tuolumne River are municipal and agricultural supply; recreation; aesthetic enjoyment; ground water recharge; fresh water replenishment; hydroelectric power generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

11. The beneficial uses of the underlying ground water are domestic, industrial, and agricultural supply.

12. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, California Code of Regulations (CCR), Section 15301.

13. Section 2511(a), Title 23, of the CCR, exempts this discharge from the requirements of Chapter 15.

14. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

15. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 79-44 is rescinded and Tuolumne Meadows Wastewater Treatment Facility, the National Park Service, and Yosemite National Park, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Effluent Limitations - Discharge to Spray Disposal Field:

1. The discharge of an effluent in excess of the following limits is prohibited:

Constituent	Units	Monthly Average	30-Day Median	Daily Maximum
20°C BOD_5	mg/l	40	--	80
Settleable Matter	ml/l	0.2	--	1.0
Coliform Organisms	MPN/100ml	--	23	500
Flow	mgd	0.12	--	0.17

B. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited, except for overflow of the shallow pond caused by spring snowmelt from a 20-year snow event.

2. The bypass or overflow of untreated or partially treated waste is prohibited.

3. Discharge of waste classified as 'hazardous' or 'designated,' as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.

C. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.

2. The discharge shall not cause degradation of any water supply.

3. The discharge shall remain within the designated disposal area at all times.

4. Collected screening, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer.

5. Pond freeboard shall be maximized prior to winter closure to provide storage capacity for snowmelt.

6. The dissolved oxygen content of holding ponds shall not be less than 1.0 mg/l for 16 hours in any 24-hour period.

7. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.

D. Provisions:

1. The Discharger may be required to submit technical reports as directed by the Executive Officer.

2. The Discharger shall comply with the attached Monitoring and Reporting Program No. 92-015, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment

and its individual paragraphs are commonly referenced as "Standard Provision(s)."

4. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.

5. In the event of any change in control or ownership of land or waste discharge facilities describe herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

6. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 24 January 1992.

William H. Crooks

WILLIAM H. CROOKS, Executive Officer

TLP 12/17/91

Attachments

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 92-015
FOR

TUOLUMNE MEADOWS WASTEWATER TREATMENT FACILITY
NATIONAL PARK SERVICE
YOSEMITE NATIONAL PARK
TUOLUMNE COUNTY

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the spray disposal field. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the outlet structure of the pond will be considered adequately composited. Time of collection of a grab sample shall be recorded. The following shall constitute the effluent monitoring program:

Constituents	Units	Type of Sample	Sampling Frequency
20°C BOD_5	mg/l	Grab	Every 2 Weeks
Settleable Matter	ml/l	Grab	Weekly
Standard Minerals[1]	mg/l	Grab	Yearly
Total Coliform Organisms	MPN/100 ml	Grab	Weekly
Flow	mgd	Cumulative	Continuous
Chlorine Residual	mg/l	Grab	Weekly

[1] Standard minerals shall include all major cations (calcium, magnesium, sodium, potassium, dissolved iron) and anions (bicarbonate, carbonate, chloride, sulfate, nitrate) and include a verification that the analysis is complete (i.e., cation/anion balance)

POND MONITORING

Samples and measurements shall be done weekly, except when inaccessible due to snow.

Constituents	Units	Frequency	Location
Dissolved Oxygen	mg/l	Weekly	Each Pond
Freeboard	feet	Weekly	Each Pond

UNDERDRAIN SYSTEM MONITORING

Samples shall be collected at the outlet of the drainage system.

Frequency	Units	Frequency
Specific Conductivity @ 25°C	μmhos/cm	Monthly
Nitrates	mg/l	Monthly

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the municipal water supply can be obtained. The following shall constitute the water supply monitoring program:

Sampling Constituents	Units	Frequency
Standard Minerals[1]	mg/l	Yearly
Specific Conductivity @ 25°C	μmhos/cm	Yearly
Total Dissolved Solids	mg/l	Yearly

1 Standard minerals shall include all major cations (calcium, magnesium, sodium, potassium, potassium, dissolved iron) and anions (bicarbonate, carbonate, chloride, sulfate, nitrate) and include a verification that the analysis is complete (i.e., cation/anion balance).

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements.

Monthly monitoring reports shall be submitted to the Regional Board by the **20th day** of the following month.

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

Upon written request of the Board, the Discharger shall submit a report to the Board by **30 January** of each year. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the Discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

The Discharger shall submit to the Board an annual Winterization Report by **30 January** of each year. The Report shall describe shutdown of campgrounds, cleaning procedures, closure of the treatment plant and collection systems, and the condition of the ponds.

The Discharger shall implement the above monitoring program as of the date of this Order.

William H. Crooks

WILLIAM H. CROOKS, Executive Officer

24 January 1992

(Date)

TLP 12/17/91

INFORMATION SHEET

TUOLUMNE MEADOWS WASTEWATER TREATMENT FACILITY
NATIONAL PARK SERVICE
YOSEMITE NATIONAL PARK
TUOLUMNE COUNTY

The Tuolumne Meadows Wastewater Treatment Facility is in the South 1/2 of Section 6, T1S, R24E, MDB&M. The elevation is approximately 8600 feet. It serves the Tuolumne Meadows Campground (800 campsites), an 80-bed lodge with a dining room, a general store, a coffee shop, a service station, a ranger station, a visitor information center, employee housing, and recreation vehicle dumpings.

The existing facilities consist of a collection system, an extended aeration treatment plant, two lined oxidation-evaporation ponds totaling approximately 4.5 acres, chlorination, and an 8-acre spray disposal field with 120 individual headers. The treatment plant is comprised of a comminutor, aeration basin, and final clarifier. A 200-ft, 24-inch pipe serves as a chlorine contact chamber for pond effluent. Fencing of the spray disposal area has not been provided due to susceptibility of snow load damage, but each individual spray header has been posted.

The treatment plant is dismantled each year to prepare for the winter season. At the beginning of each season, after the snow melts, the plant is reassembled and put into operation. Large amounts of infiltration, cold water temperatures, and a short operational season minimize the treatment efficiency of the plant.

The liner in the deeper pond was replaced in September 1991 because the old liner had a tear and bubbled up from springs beneath the pond. The new liner has an underdrain system which will drain to Tuolumne Meadows. There will be monitoring requirements on this drainage.

The shallow pond may overflow in a 1-in-20-year snow event in the spring when the snow melts. However, since the ponds are drawn down prior to winter, the overflow would consist mostly of snowmelt. Additionally, flow in the Tuolumne River (the receiving water) would be high from the spring runoff, providing high dilution to overflow.

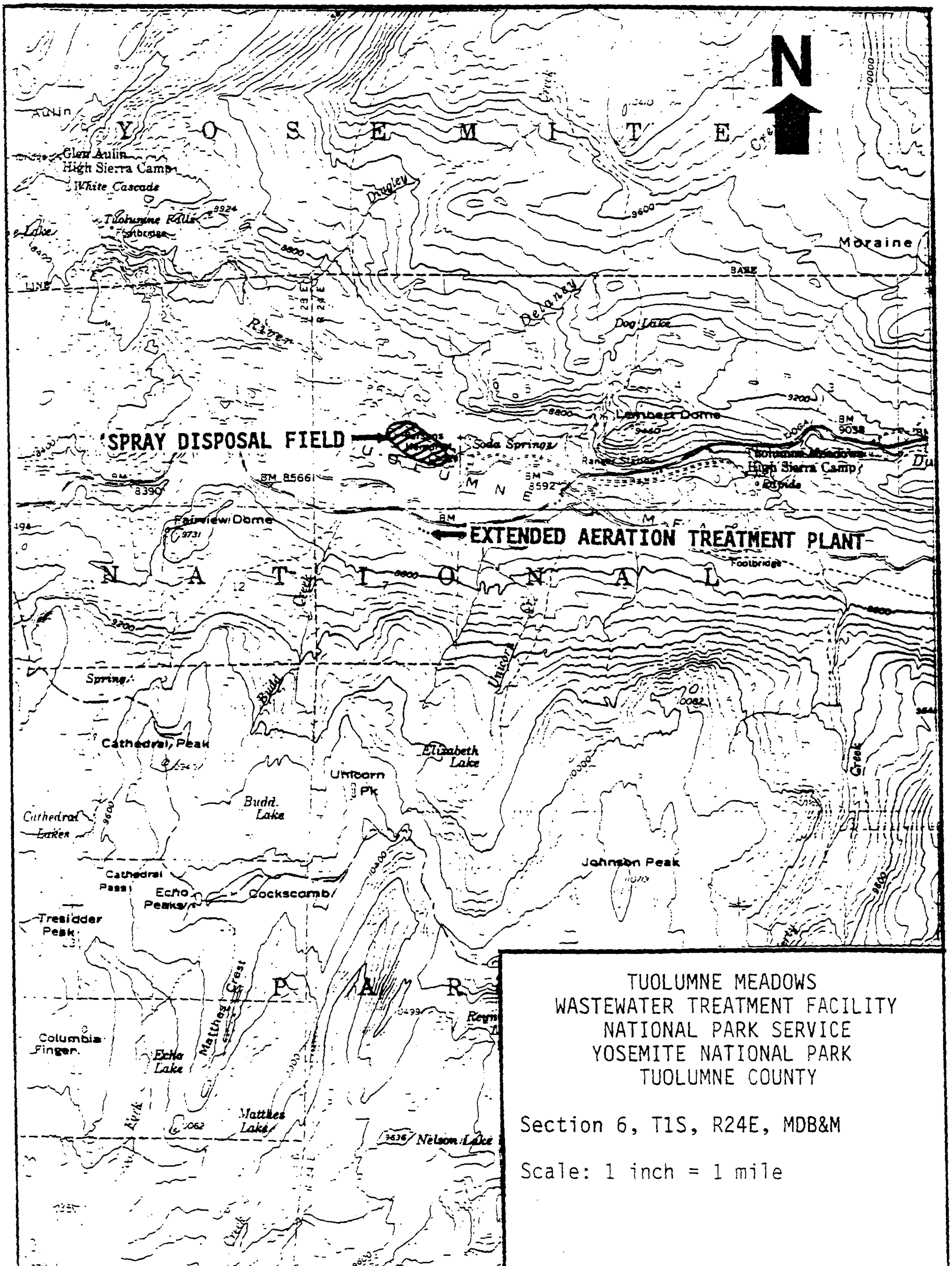
The sewer collection system is subject to large volumes of infiltration, estimated to be as high as 90,000 gpd from flooding and high ground water conditions during the winter, spring, and early summer. After the treatment plant is dismantled in preparation for the winter, infiltration water in the collection system is discharged through a drain line to Tuolumne Meadows.

Ground water information in the Tuolumne Meadows area is not readily available due to the absence of water wells. Soils consist of decomposed granite and glacial moraine underlain by solid granite bedrock. Numerous rock outcroppings suggest that soil depth is variable and shallow. Surface waters in the area originate as the snow melts.

Beneficial uses of surface water in the Tuolumne Meadows area are recreational and include fishing, aesthetics, and water supply for campers and hikers in the area. Downstream water uses include domestic water supply from Hetch Hetchy Reservoir which serves the City of San Francisco and other Bay area communities, as well as recreation including body contact activities, freshwater and wildlife habitat, irrigation, stock watering, and hydropower production.

Annual precipitation is approximately 33 inches, which includes winter snow accumulations as high as 16 to 20 feet.

TLP



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Section 6, T1S, R24E, MDB&M

Scale: 1 inch = 1 mile